

Anahita Bolourani

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Education

2024 – Present **University of California, Los Angeles (UCLA)**

Master of Science – Statistics and Data Science

2022 – 2026 **University of California, Los Angeles (UCLA)**

Doctor of Philosophy – Engineering

Publications

Banayeeanzade, A., Tak, A. N., Bahrani, F., Bolourani, A., Blas, L., Ferrara, E., Gratch, J., & Karimireddy, S. p. "Psychological Steering in LLMs: An Evaluation of Effectiveness and Trustworthiness", *arXiv preprint*, 2025.

Tak, A. N., Bolourani, A., Shank, D. B., & Gratch, J. "Impact of LLM Alignment on Impression Formation in Social Interactions", *COLM*, 2025.

Tak, A. N., Banayeeanzade, A., Bolourani, A., Kian, M., Jia, R., & Gratch, J. "Mechanistic Interpretability of Emotion Inference in Large Language Models," *Findings of ACL*, 2025.

Bolourani, A. Burton, H. Bozorgnia, Y. "Evaluating Procedures for Estimating Vertical Seismic Load Effects Specified in the U.S. Building Code," *ASCE (JSE)*, 2024.

Bolourani, A. et al. "Structural health monitoring of using support vector machine and principal component analysis," *Structures*, 2021.

Tak, A. N., Bolourani, A. et al. "BIM-based 4D mobile crane simulation and onsite operation management," *Automation in Construction*, 2021.

Bolourani, A., Tak, A. et al. "Developing a BIM-SHM Integrated System for Disaster Risk Management Using Auto-Regressive Model," 2019.

Research Experience

2025–Present **Researcher, TrustworthyAI Lab, UCLA**

- Fine-tuned *TabSyn* diffusion models on heterogeneous tabular datasets with reinforcement learning (course Project)

2024–Present **Independent Project**

- Led a study on how alignment shapes social-impression formation in LLMs

2022–2025 **Graduate Research Assistant, B. JohnGarrick Institute, UCLA**

- Engineered a Python analytics platform that streams Community Seismic Network and NASA JPL strong-motion sensor data, using fast Fourier transforms to extract and track dominant structural frequencies
- Performed statistical analysis on multimillion-record structural-response datasets, informing proposed updates to U.S. building-code vertical-load criteria.

2019–2021 **Research Assistant, University of Tehran**

- Designed an SVM-PCA SHM model to detect structural damage
- Applied SOM (Self-Organizing Maps) to benchmark structures, demonstrating robust anomaly detection (unsupervised learning) under noisy conditions.

Technical Skills

Programming: Python (Pandas, NumPy, PyTorch, SciPy, scikit-learn, ObsPy, Matplotlib), R

ML Tooling: Large Language Models (LLM), Diffusion Models, Machine Learning (ML)

Data/Dev Ops: Git, LaTeX, Docker

Selected Coursework

STATS M231A Pattern Recognition & Machine Learning,
STATS C261 Pattern Recognition & Machine Learning
STATS 200A Applied probability,
STATS 200B Theoretical Statistics,
STATS 201A Research Design & Sampling,
STATS 202A Statistical Programming
STATS C216 Applied Bayesian Social Statistics,
STATS 213 Synthetic Data Generation,

Certificates

“Statistical Learning,” Stanford Online
Google Data Analytics Professional Certificate (4 courses)
“Neural Networks and Deep Learning,” DeepLearning.AI

Awards & Honors

ASCE Metropolitan Los Angeles *Le Val Lund Memorial Award*, 2024
ASCE Los Angeles Section *Outstanding Graduate Student Scholarship*, 2023 & 2024
UCLA *Summer Mentored Research Fellowship*, 2022
UCLA *Graduate Student Researcher Award*, 2022
Semi-finalist, Mathematics National Olympiad (Iran), 2011